

Telecom Customer Churn Analysis

Power BI Dashboard • Project Report

File analyzed	Dashboard.pbix
Report pages in dashboard	4 (EDA, Dashboard, Customers, Churn Analysis)
Source dataset	Telco Customer Churn — customer-level records with churn probability & risk scoring
Total customer records	7,043

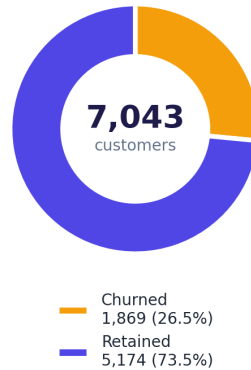
Dashboard flow — four report pages



1. Executive Summary

This report documents a Power BI dashboard built on a telecom customer churn dataset. The workbook analyzes churn behavior across 7,043 customers, of whom 1,869 (26.53%) have churned, and augments the raw customer data with a predictive Churn Probability and Risk Score to flag at-risk accounts. The dashboard moves from exploratory analysis, to an executive summary view, to a per-customer lookup, and finally to a risk-focused action page.

Overall Churn Split



2. Project Objective

To understand which customer attributes (tenure, contract type, internet service, payment method, add-on services, and demographics) are associated with churn, and to surface a prioritized list of high-risk customers so the business can act before they leave.

3. Dataset & Data Model

The model is built on a single fact table (named in the file as *2-Telco-Customer-Churn-prob*), indicating the source data was pre-scored with a churn probability model before being loaded into Power BI. Key fields used across the report:

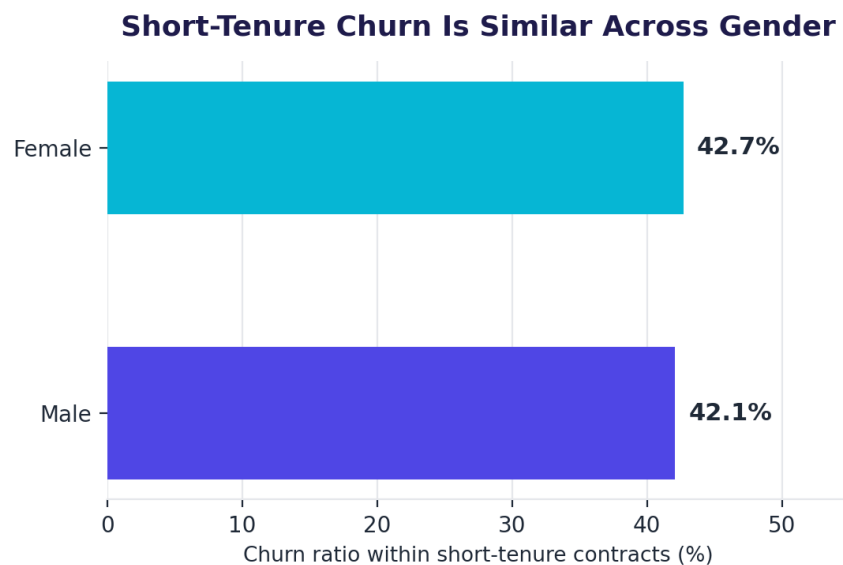
- customerID, gender, SeniorCitizen, Partner — customer identity & demographics
- tenure, Contract, PaperlessBilling, PaymentMethod — account & billing behavior
- PhoneService, InternetService, OnlineSecurity, DeviceProtection, TechSupport — services subscribed
- TotalCharges — cumulative revenue per customer
- Churn — actual churn flag (Yes/No)
- Churn_Prob, Risk — model-derived churn probability and risk classification

4. Dashboard Structure & Findings

Page 1 — EDA (Exploratory Data Analysis)

Univariate and bivariate breakdowns of churn using column, clustered-column and area charts, plus summary cards for total records and churn count.

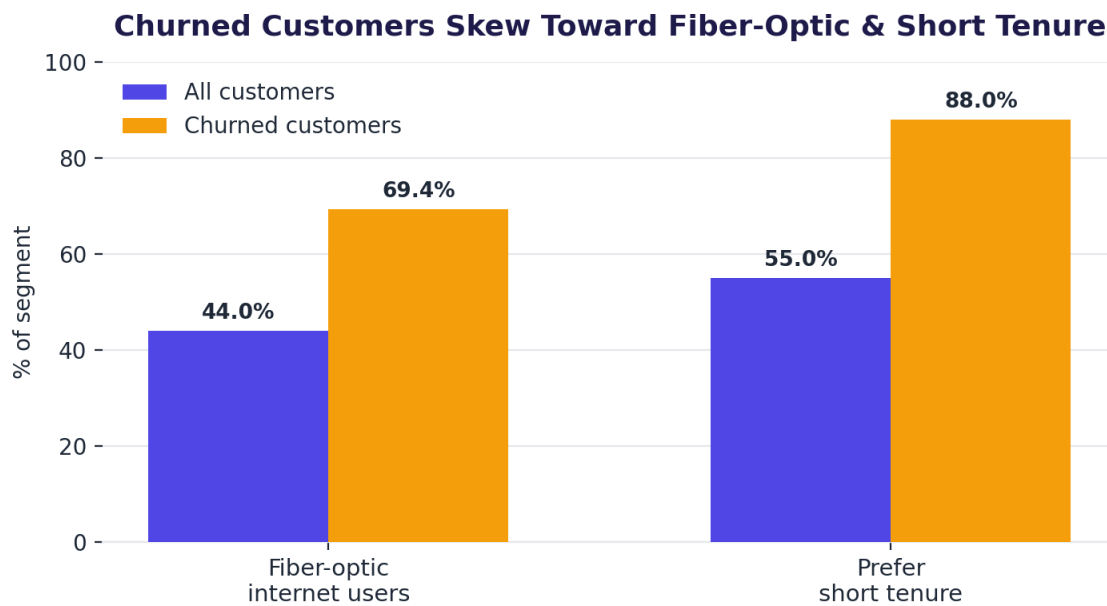
- Gender has an approximately equal churn ratio between male and female customers.
- Longer-tenure contracts show a lower churn ratio than shorter-tenure contracts.
- Senior citizens churn at close to 50%, roughly double the overall rate.
- Of 7,043 total customers, 1,869 have churned (26.53%).
- The area chart shows churn rate declining steadily as tenure increases.



Page 2 — Dashboard (Executive Summary)

A side-by-side comparison of the full customer base ('Customer Profile') against the churned subset ('Churner Profile'), built from donut, pie, clustered bar/column charts and KPI cards.

- Customer Profile: ~6,400 of 7,043 customers use phone service; most lack online security or tech support; 43.96% use fiber-optic internet.
- Customer Profile: 55.02% of customers choose shorter tenure plans; electronic check is the most common payment method.
- Churner Profile: of 1,869 churned customers, ~1,700 had phone service, ~1,500 had no online security, and ~1,400 had no tech support; 69.4% used fiber-optic internet.
- Churner Profile: 88.05% of churned customers had chosen shorter tenure plans, and electronic check was again the most common payment method.



Page 3 — Customers

A searchable, per-customer lookup view with a slicer and KPI cards showing an individual customer's personal info, phone service, contract, tenure, preferred payment mode, risk level and churn probability — built for account-level drill-down rather than aggregate analysis.

Page 4 — Churn Analysis

A risk-focused action page combining KPI cards with a bar and column chart to quantify the at-risk population and its financial implication.

- 2,900 of the 7,043 customers are classified as risky and need action.
- The average Risk Score across customers is 33.49, which the analysis flags for reduction to at least half.
- The dashboard warns of major revenue loss if the at-risk segment is not retained.

Customer Risk Segmentation



5. Key Findings Summary

Metric	Value
Total customers	7,043
Churned customers	1,869 (26.53%)
Customers flagged high-risk	2,900 (41.2%)
Average risk score	33.49
Short-tenure preference — all vs. churned	55.02% → 88.05%
Fiber-optic internet — all vs. churned	43.96% → 69.4%
Top service gap among churners	No online security / no tech support
Most common payment method	Electronic check

6. Tools & Techniques Used

- Power BI Desktop for data modeling, DAX-driven KPI cards, and report layout across 4 pages.
- A pre-computed churn probability / risk score field, implying a predictive model was run upstream (e.g. in Python or Excel) and its output merged into the customer table.
- A mix of visual types — column, clustered column, clustered bar, donut, pie, area, and card visuals — chosen to match univariate, bivariate, and KPI-style storytelling.
- A slicer-driven customer lookup page for operational, record-level use by account teams.

7. Recommendations

- Prioritize retention outreach to the 2,900 customers flagged as high-risk, starting with those on short-tenure, fiber-optic, electronic-check profiles — the segment most over-represented among past churners.
- Bundle online security and tech support into onboarding offers, since their absence tracks closely with churn.
- Incentivize longer-tenure contracts (annual/two-year) for new and at-risk customers, since tenure is the strongest inverse driver of churn observed in the EDA page.
- Set a target to cut the average risk score from 33.49 toward half that value, and track it as a recurring KPI on the Dashboard page.